



COMP4141 Slides Week 4

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Admin

Tests

- Marks will be on formatif
- Test is closed book (just to clarify)
- You will not be getting you sheet back but you can ask if you had questions
- For week 3 (last week) some lost 1 mark for having a solution that doesn't use the ϵ -NFA construction like the question asks for.
- The **week-5** (next week) in-class assessment will be to use the context-free pumping lemma to prove some language to be non-context-free.

Marking

For week 1:

- The general patterns were most did not reference/justify why $|aa| = 2$ or $|x.y| = |x| + |y|$.
- Some used an alternative definition such as ax instead of xa .
- Some used induction on the length of the string directly instead of a structural induction.

For week 2:

- Pass and Credit tasks were generally well done by those who attempted them.
- For the Distinction task, many students had overly long or complicated proofs. Try to use structured induction (I recommend using the tutorial solution as a guide).
- HD task was correct on the first attempt (the counterexample was simple).

More on Marking

- Consider lowering your target grade (or raising it) if necessary!

Marking

Apologies for any confusion before

- Those who submit before the due date get feedback. The feedback comes in one week from me, and then you will have another week to take the feedback and revise. Together, this is a three week process, and two weeks after the first due date.
- You can still submit after the due date (first due date), and before the deadline (final due date).

Properties of CFLs

Pumping Lemma for CFLs

Pumping Lemma

If $L \subseteq \Sigma^*$ is context-free, then there exists $p \in \mathbb{N}$ (the pumping length) such that if $w \in L$ with $|w| \geq p$, then w can be written as

case
free

satisfying

- $|vy| > 0$,
- $|vxy| \leq p$,
- $uv^ixy^iz \in L$ for all $i \in \mathbb{N}$.

$$w = uvxyz$$

Pump up
 uv^2xy^2z

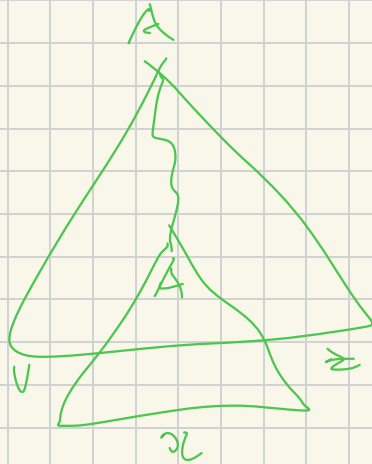
$uvxy^2z$

Pump down
 uxz

$uvvxy^2z$

uxz

Pumping Lemma Intuition



CFL Algorithms

Chomsky Normal Form

S is the starting symbol

Chomsky Normal Form

A context-free grammar is in **Chomsky normal form (CNF)** if every production is of one of the forms

$$S \rightarrow \epsilon, \quad A \rightarrow BC, \quad A \rightarrow a,$$

where B, C are variables (and neither is S) and $a \in \Sigma$.

Theorem

Every context-free language is generated by a CFG in Chomsky normal form.

$A \rightarrow \underline{d}B\underline{c}C$ not allowed

Converting to Chomsky Normal Form

Algorithm

1. **Add a new start variable.** Introduce a fresh start symbol S_0 with production $S_0 \rightarrow S$. Remove ϵ -productions (except possibly from S_0).
2. **Remove terminals from long RHS.** Replace terminals appearing with variables by new variables X_a and add productions $X_a \rightarrow a$.
3. **Remove unit productions** ($A \rightarrow B$). Replace them by the productions of B .
4. **Reduce RHS to pairs of variables.** Replace productions of the form

$$A \rightarrow B_1 B_2 \cdots B_n \quad (n > 2)$$

with

$$A \rightarrow B_1 C_1, C_1 \rightarrow B_2 C_2, \dots, C_{n-2} \rightarrow B_{n-1} B_n$$

using fresh variables C_i .

12/22

$$N_{ij} \rightarrow N_{ik} N_{k+1,j}$$

$w \in a_1 \dots a_i \dots a_j \dots a_n$

CYK Algorithm

CYK Algorithm

Let $G = (V, \Sigma, P, S)$ be a CFG in Chomsky normal form and let $w = a_1 \cdots a_n$.

1. **Base case.** For $i = 1, \dots, n$ set

$$N_{ii} = \{A \in V : A \rightarrow a_i \in P\}.$$

2. **Build larger substrings.** For $d = 1, \dots, n-1$ and $i = 1, \dots, n-d$, let $j = i+d$ and compute

$$N_{ij} = \{A \in V : A \rightarrow BC \in P, B \in N_{ik}, C \in N_{k+1,j} \text{ for some } i \leq k < j\}.$$

3. **Accept if** $S \in N_{1n}$.

W4 P1

Problem 1

Show that the following languages are not context-free:

(a) $\{0^n 1^n 0^n 1^n : n \in \mathbb{N}\}$

$\uparrow \uparrow$
L

Suppose that L is context free, then $\exists p$

s.t. $\forall |w| \geq p \quad w = uvxyz:$

① $|vy| > 0$

② $|vxy| \leq p$

③ $uv^i xy^i z \in L \quad \forall i \in \mathbb{N}$

Try $W = 0^p 1^p 0^p 1^p$

If $W = uvxyz \in$

By 2 $|vxyz| \leq p$ 2 3 Group 4



Group 1

Not possible i.e. can be $0^a 1^b 0^c$ or $1^a 0^b 1^c$

So $|xy|$ is in at most 2 groups

$$u \succ z$$

* Then since $|vy| > 0$
 $|v| > 0$ or $|y| > 0$ so
at least one block
decreases in size but
at most

$$u v^2 \succ (y^2) z$$

In this case
we have extra 0 's
if's in no more than
 2 group. Since $|v| > 0$
or $|y| > 0$ at least,
one group does include

Example

$p \vee a$ $p \vee b$ p p
 $\neg$ $\neg$ $\neg$ $\neg$
where $a > 0$ or $b > 0$

When

p p p $\neg$
 $\neg$ $\neg$ $\neg$ $\neg$

 $\neg$ $\neg$ $\neg$ $\neg$

Can use either
case " We get
our contradiction!

$$\Sigma = \{0, 1, \#\}$$

Problem 1

Show that the following languages are not context-free:

(b) $\{w\#t : w, t \in \{0, 1\}^* \text{ and } w \text{ is a substring of } t\}$

Assume L is context free and suppose p is given

Look at $|w| \geq p$ where $w = 0^p \# 0^p$

In L : examples

$010 \# 0110$
 $\underbrace{\quad\quad\quad}_w \quad \underbrace{\quad\quad\quad}_+$

$0001 \# 000110$ is in L ?
 $\underbrace{\quad\quad\quad}_w \quad \underbrace{\quad\quad\quad}_+$

$Q^p \# Q^p$ does not work

since we could have

$$u = Q^{p-k}$$

$$v = Q^k$$

$$x = \#$$

$$y = Q^k$$

$$z = Q^{p-k}$$

$$uv^i x y^i z$$

$$= Q^{p-k} \# Q^k \# Q^{p-k}$$

Side 1

Side 2

Try $0^p 1^p \neq 0^p 1^p$

$w = uvxyz$

$|vxy| \leq p$

Case 1: v contains $\#$
or y contains $\#$

Pump up or pump down we
don't have exactly one $\#$
as needed

Case 2: v and y are on
different sides

So wxy is in $1^p \neq 0^p$
 $\uparrow \quad \downarrow$
 v is 1^a y is 0^b

If we pump up

$0^p 1^{p+2a}$ $\neq$ $0^{p+2b} 1^p$

↓ This is in L iff $a=0$

So pumping down

So $b \neq 0$

$$u x y = 0^p 1^p \neq 0^{p-b} 1^p \notin L$$

Case 3(a) v and y are
both on side z

then we pump down
and have

$$O^a \nsubseteq O^b \nsubseteq O^{a-b} \nsubseteq L$$

Where $a > 0$ or $b > 0$

(Case 2(b)) v and y are
on side $\perp$ then we
pump

Problem 1

Show that the following languages are not context-free:

(c) $\{0^{2^n} : n \in \mathbb{N}\}$

Assume L is context free and we
are given p , try 0^{2^p}

We know that $|vocy| \leq p$
and $|vy| > 0$

$$Vy = 0^k \text{ where}$$

$$0 < k \leq p$$

When we pump up
we have

$$a) \quad 0^{z^p + k}$$

For ⑨ to be in the language we need

$$2^p + k = 2^n$$

$$2^p + k = 2^n$$

Since $0 < k$, and $k \leq p$

So since $p < 2^p$

$$\underline{2^p} < 2^p + k \leq 2^p + p < 2^p + 2^0 < \underline{2^{p+1}}$$

So $2^p + k$ is not a power
of 2

W4 P2

Problem 2 (a)

Solve the following problems by first converting the grammar to Chomsky Normal Form, if necessary, and then applying the CYK algorithm.

(a) Let G_1 be the grammar $(\{S, B\}, \{a, b, c\}, R, S)$ where R consists of the following rules:

$$S \rightarrow aSc \mid ac \mid B$$

$$B \rightarrow \epsilon \mid Bb$$

Decide if $abbc \in L(G_1)$

Convert to CNF form

Step 1
(add S_0)

$$S_0 \rightarrow S$$

$$S \rightarrow aSc \mid ac \mid B$$

$$B \rightarrow \epsilon \mid Bb$$

Then remove ϵ

$$S_0 \rightarrow S$$

$$S \rightarrow aSc \mid ac \mid B$$

$$B \rightarrow b \mid Bb$$

$$S_0 \rightarrow S \mid \epsilon$$

$$S \rightarrow ac \mid aSc \mid B$$

$$B \rightarrow b \mid Bb$$

Step 2

Remove
terminals

$$S_0 \rightarrow S \mid \varepsilon$$

$$S \rightarrow X_a S X_c \mid X_a X_c \mid B$$

$$B \rightarrow b \mid B X_b$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

$$X_c \rightarrow c$$

Step 3

Remove
 $A \rightarrow B$

$$S_0 \rightarrow S \mid \varepsilon$$

$$S \rightarrow X_a S X_c \mid X_a X_b \mid b \mid B X_b$$

$$B \rightarrow b \mid B X_b$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

$$X_c \rightarrow c$$

Step 4

Remove

$A \rightarrow B_1 B_2 \dots B_k$

$$S_0 \rightarrow S \mid \varepsilon$$

$$S \rightarrow S_1 X_c \mid X_a X_c \mid b \mid B X_b$$

$$S_1 \rightarrow X_a S$$

$$B \rightarrow b \mid B X_b$$

$$X_a \rightarrow a$$

$$X_b \rightarrow b$$

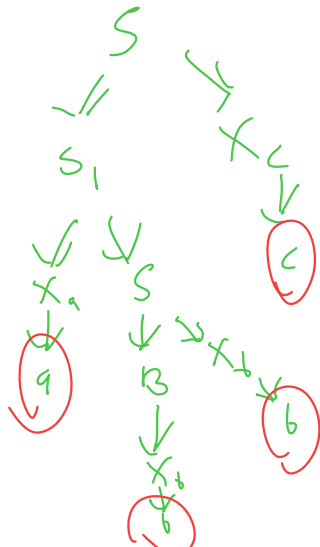
$$X_c \rightarrow c$$

Problem 2 (a) CYK Table



S			
S_1			
S_1	S	S_1	
X_a	$S B$ X_b	$S B$ X_b	X_c
a	b	b	c

$abbc$



Problem 2 (b)

Solve the following problems by first converting the grammar to Chomsky Normal Form, if necessary, and then applying the CYK algorithm.

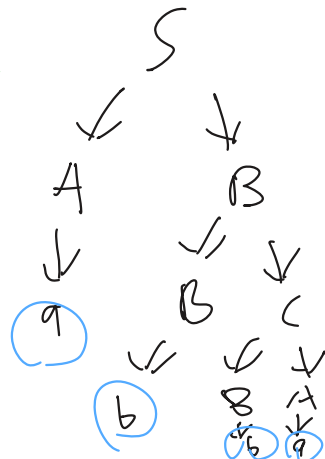
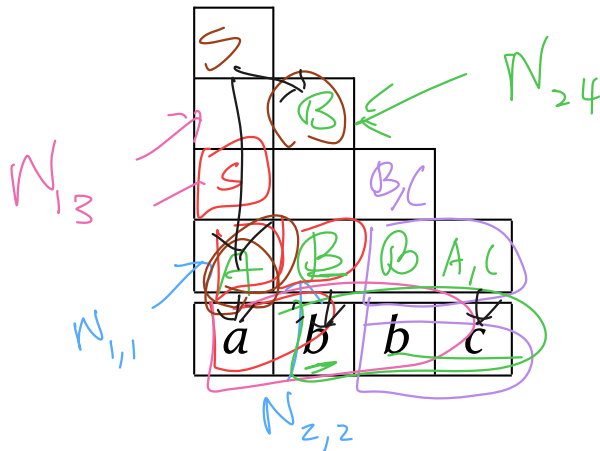
- (b) Let G_2 be the grammar $(\{S, A, B, C\}, \{a, b, c\}, R, S)$ where R consists of the following rules:

$$\begin{array}{lcl} \underline{S} & \rightarrow & AB \\ A & \rightarrow & CC \mid \underline{a} \mid c \\ \underline{B} & \rightarrow & \underline{BC} \mid \underline{b} \\ C & \rightarrow & \underline{CB} \mid \underline{BA} \mid \underline{c} \end{array}$$

Decide if abbc $\in L(G_2)$

$$abbc \in L$$

Problem 2 (b) CYK Table



Problem 2 (c)

Solve the following problems by first converting the grammar to Chomsky Normal Form, if necessary, and then applying the CYK algorithm.

- (c) Let G_3 be the grammar $(\{S, A, B, C\}, \{a, b, c\}, R, S)$ where R consists of the following rules:

$$S \rightarrow ASB \mid \epsilon$$

$$A \rightarrow C \mid BB \mid a$$

$$B \rightarrow BCC \mid b \mid \epsilon$$

$$C \rightarrow CBA \mid BAC \mid c$$

Decide if $abbc \in L(G_3)$

(convert to CNF form)

Step 1

Add S_0
and remove ϵ

$$S_0 \rightarrow S \mid \epsilon$$

$$S \rightarrow ASB \mid AB$$

$$A \rightarrow C \mid BB \mid a$$

$$B \rightarrow BCC \mid b \mid \epsilon$$

$$C \rightarrow CBA \mid BAC \mid C$$

$$S_0 \rightarrow S \mid \epsilon$$

$$S \rightarrow AB \mid ASB \mid AS \mid A$$

$$A \rightarrow C \mid BB \mid a \mid B$$

$$B \rightarrow BCC \mid b \mid CC \mid C$$

$$C \rightarrow CBA \mid BAC \mid C \mid AB \mid A \mid B$$

Step 2

Remove
 $X \rightarrow a \dots Y$

Nothing to do

Step 3

Remove
 $A \rightarrow B$ rules

$S_0 \rightarrow ASA \mid AA \mid AS \mid AAA \mid a \mid b \mid c \mid c$

$S \rightarrow ASA \mid AA \mid AS \mid AAA \mid a \mid b \mid c$

$A \rightarrow AA \mid AA \mid a \mid b \mid c$

(B and C are the same)

Step 4

Remove

3+ step

$$S_0 \rightarrow S_1 A \mid A A \mid A S \mid S_2 A \mid a \mid b \mid c$$

$$S \rightarrow S_1 A \mid A A \mid A_2 S \mid S_2 A \mid a \mid b \mid c$$

$$A \rightarrow A A \mid S_2 A \mid a \mid b \mid c$$

$$S_1 \rightarrow A S$$

$$S_2 \rightarrow A A$$

Problem 2 (c) CYK Table

S *			
*	*		
*	*	*	
A, S, S_0	A, S, S_0	A, S, S_0	A, S, S_0
a	b	b	c

Another example

$$S \rightarrow ASB$$

↓

$$A \in B$$

↓

$$aBAA$$

↓

$$abBB \rightarrow ab \epsilon \epsilon bc = abbc$$

$$S \rightarrow ASB \mid \epsilon$$

$$A \rightarrow BB \mid a \mid bc$$

$$B \rightarrow BAA \mid b \mid \epsilon$$

Step 1:

Adding S_0

and removing ϵ

$$S_0 \rightarrow S$$

$$S \rightarrow AS(B) | \epsilon$$

$$A \rightarrow (B)(B) | a | bc$$

$$B \rightarrow (B)AA | b | \epsilon$$

$$S_0 \rightarrow (S)$$

$$S \rightarrow A(S)B | \epsilon | A(S)$$

$$A \rightarrow BB | a | bc | B$$

$$B \rightarrow BAA | b | AA$$

$$S_0 \rightarrow S \mid \varepsilon$$

$$S \rightarrow ASB \mid AS \mid A \mid AB$$

$$A \rightarrow BB \mid a \mid bc \mid B$$

$$B \rightarrow BAA \mid b \mid AA$$

Step 2

Remove

$$X \rightarrow a \dots y$$

$$S_0 \rightarrow S \mid \varepsilon$$

$$S \rightarrow ASB \mid AS \mid A \mid AB$$

$$A \rightarrow BB \mid a \mid x_a x_c \mid B$$

$$B \rightarrow BAA \mid b \mid AA$$

$$x_a \rightarrow a$$

$$x_c \rightarrow c$$

Step 3

Remove

$A \rightarrow B$

$S_0 \rightarrow S \mid \varepsilon$

$S \rightarrow ASB \mid AS \mid A \mid AB$

$A \rightarrow BB \mid a \mid x_b x_c \mid BAA \mid b \mid AA$

$B \rightarrow BAA \mid b \mid AA$

$S_0 \rightarrow ASB \mid AS \mid AB \mid BB \mid a \mid x_b x_c \mid BAA \mid b \mid AA$

$S \rightarrow ASB \mid AS \mid AB \mid BB \mid a \mid x_b x_c \mid BAA \mid b \mid AA$

$A \rightarrow BB \mid a \mid x_b x_c \mid BAA \mid b \mid AA$

$B \rightarrow BAA \mid b \mid AA$

Step 5

Remove

$A \rightarrow B_1 B_2 \dots B_n$

$S_0 \rightarrow S_1 B | A S | A B | B B | a | X_b X_c | S_2 A | b | A A$

$S \rightarrow S_1 B | A S | B B | A B | a | X_b X_c | S_2 A | b | A A$

$A \rightarrow B B | a | X_b X_c | S_2 A | b | A A$

$B \rightarrow S_2 A | b | A A$

More CYK examples

$S \rightarrow AB | \epsilon$

$A \rightarrow BB | a | c$

$B \rightarrow BA | b$

	A			
	A	B		
A	B	B	A	
a	b	b	a	

abba not
in this
one!

$S \rightarrow \cancel{B}A \rightarrow ABA \rightarrow ABB\cancel{A}$
 $\downarrow$
a b b c

$S \rightarrow BA | \epsilon$

$A \rightarrow BA | a | c$

$B \rightarrow AB | b$

S				
A				
B	A	A	S	
A	B	B	A	
a	b	b	a	

Feedback

Please let me know how I can improve for next week [here](#)!

COMP4141 26T1 Feedback Form
(For Brendan Mabbutt's Tutorials)

